

CS 217: Practice questions on Generative Classifiers

1. Consider a binary classification problem ($y \in \{0, 1\}$) and two dimension data ($d = 2$) where both x_1 and x_2 are binary. Let $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$ denote the number of training instances in class y where $[x_1 \ x_2]$ is $[0 \ 0]$, $[1 \ 0]$, $[0 \ 1]$, $[1 \ 1]$ respectively.

(a) Suppose we use a naive Bayes classifier where x_1 is assumed independent of x_2 .

- i. What is the maximum likelihood estimate of the two Bernoulli parameters in class y in terms of the counts $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$

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- ii. What is the maximum likelihood estimate of $\Pr(y)$ in terms of the counts $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$

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- iii. For what values of $n_{00}^y, n_{10}^y, n_{01}^y, n_{11}^y$ will the naive Bayes classifier incur the maximum percentage training error? Justify. Assume number of instances in both classes is the same.

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- (b) Now suppose we decide to use a model more powerful than naive Bayes. In each class, we model the probability that $\Pr([x_1 \ x_2] = [a \ b] | y)$ with a parameter p_{ab}^y . Further we share some parameters between the two classes to reduce the number of parameters by requiring that $p_{11}^0 = p_{11}^1 = p_{11}$. Clearly, $p_{00}^y + p_{10}^y + p_{01}^y + p_{11} = 1$. Write the log-likelihood function on the training data in terms of the counts n_{ab}^y , and parameters p_{ab}^y , $\Pr(y = 1)$, $\Pr(y = 0)$.

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- (c) Derive the maximum likelihood estimates of the $p_{00}^y, p_{10}^y, p_{01}^y$ parameters only in terms of p_{11} and the training counts. [Hint: Use Lagrangian multiplier to push the two constraints $p_{00}^y + p_{10}^y + p_{01}^y + p_{11} = 1$, one for each y to the objective.]

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(d) Now derive the maximum likelihood estimate for p_{11} .

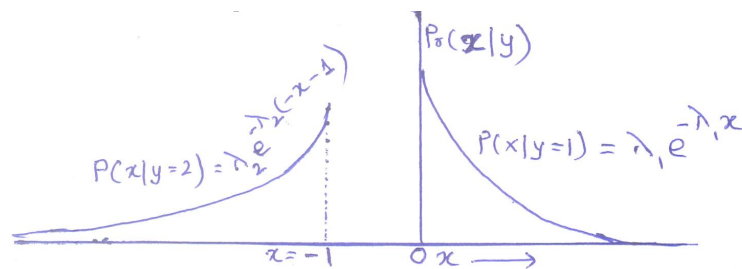
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2. The exponential density function is a popular parametric density function to capture the distribution of positive reals z . It is defined as:

$$\begin{aligned}\exp(z; \lambda) &= \lambda e^{-\lambda z} \quad \text{if } z \geq 0 \\ &= 0 \quad \text{otherwise.}\end{aligned}$$

The mean, variance, cumulative density of the above are $\lambda^{-1}, \lambda^{-2}, 1 - e^{-\lambda z}$ respectively.

Suppose you apply the naive Bayes classifier on a dataset where the *true* $\Pr(x|y)$ distribution is defined as $\Pr(x|y = 1) = \exp(x; \lambda_1)$ and $\Pr(x|y = 2) = \exp(-1 - x; \lambda_2)$. Assume $\lambda_1 > 0, \lambda_2 > 0$. The picture below shows an example distribution for the two classes with different λ_1, λ_2 .



The true values of λ_1, λ_2 is not known to us, nor do we know that the data was generated from this special density function. We are only given a training set D by sampling a large number $N/2$ of instances from each of the two classes. When training the naive Bayes classifier, we will continue to use a Gaussian distribution to model $P(x_1|y)$ for each of the two classes. Thus, we will need to train parameters mean μ_y and variance σ_y^2 for each y .

- (a) Write down the maximum likelihood estimates of the following parameters (in terms of λ_1, λ_2) assuming N is very large. The first question is solved for you to give you an idea of the solution:

i. μ_1
Answer: λ_1^{-1}

ii. μ_2

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iii. σ_1^2 , and σ_2^2

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(b) For what value of w_2, w_1, w_0 is predicting class 1 based on the hyperplane $w_2x^2 + w_1x + w_0 > 0$, the same as the predictions made by the above naive Bayes classifier.

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(c) For what value of λ_1, λ_2 is the training error minimum? What is the value of the minimum error? Justify.

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